When Perplexity Lies: A Controlled Perplexity Collapse with No Downstream Degradation in Weight-Quantized LLMs

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Abstract

Post-training quantization of large language models is evaluated overwhelmingly by WikiText perplexity, and perplexity regressions are routinely read as utility loss. We show this inference can fail badly. Using outlier-channel protection—keeping a small fraction of input channels in FP16 on a low-bit base—we construct a controlled generator of extreme perplexity damage: on an error-compensated GPTQ-4 base, a selector that ranks channels by the Hessian-weighted residual objective raises Llama-3.2-3B WikiText-2 perplexity from 8.17 to 51.68 at matched weight-only storage. Disk reload verifies the evaluated checkpoint is that model (51.76). Yet on six zero-shot tasks it loses nothing: macro accuracy 67.61% versus 67.10% for its own base, a paired-bootstrap difference of $+0.51$ points whose interval excludes zero, and LAMBADA perplexity 4.17. The pattern replicates on Llama-3.2-1B (10.56 to 63.83 reload-verified, $+0.32$ points). A per-token decomposition rules out the obvious explanation: the damage is broad, not a tail artifact—74.6% of tokens degrade, the median token worsens, and excluding the worst 5% still leaves a $4\times$ gap. We further show the damage requires the selector's off-diagonal Hessian coupling, and report that the natural mechanism, protection destroying error compensation, is falsified by direct measurement.

1 Introduction

Weight-only post-training quantization (PTQ) is the default way to shrink large language models, and the field's default report is a WikiText perplexity table. Methods are accepted or rejected on perplexity deltas that are often far smaller than one point, and deployment gates are frequently written in the same terms. Implicit in this practice is an assumption: that perplexity movement tracks the model's usefulness.

We show that assumption can fail by a wide margin, using a construction entirely within standard PTQ practice. *Outlier-channel protection*—keeping a small fraction of input channels in FP16 on top of a low-bit base—is a common recipe [Dettmers et al., 2022, Lee et al., 2024, Dettmers et al., 2023]. Its selection signal is usually presented as a design detail. We find that one standard choice of signal, applied to an error-compensated base, produces a catastrophic perplexity collapse: on Llama-3.2-3B with a GPTQ-4 base [Frantar et al., 2023], protecting 2% of input channels chosen by the Hessian-weighted residual objective moves WikiText-2 perplexity from **8.17 to 51.68**, at matched theoretical weight-only storage, while random selection of the same number of channels leaves it at 8.18.

A $6.3\times$ perplexity increase would normally end the discussion. We instead treat it as an instrument. We export that exact configuration to a dense checkpoint, verify by in-memory materialization and by disk reload that the checkpoint *is* the perplexity-51.76 model, and then evaluate it on six standard zero-shot tasks. It shows no degradation at all: macro accuracy **67.61%** against **67.10%** for its own unprotected base and **68.43%** for FP16, a paired bootstrap difference of $+0.51$ **points** $[+0.13, +0.86]$— statistically *favourable*— with LAMBADA perplexity **4.17**, comparable to the base's 4.28.

Contributions.

1. A reload-verified demonstration that a $\sim 6\times$ WikiText-2 perplexity collapse can carry **zero downstream cost** on six tasks, on **two models**, including a token-level task on a second corpus (§4).
2. A **per-token decomposition falsifying the tail explanation**: the degradation is broad (74.6% of tokens worse, median ΔNLL $+0.287$), so the decoupling is not an artifact of a few catastrophic tokens (§4.3).
3. A characterization of *what generates* the perplexity damage: it requires the selector's **off-diagonal Hessian coupling**. Selectors using no Hessian, only its diagonal, only residual magnitude, or random selection are all benign at every budget we tested (§5).
4. A **falsified mechanism**, reported as such: the natural explanation—that protection destroys the base's error compensation—predicts that harmful selectors target compensation-bearing columns. Direct measurement shows the opposite ordering (§6).
5. A quantification of how much the effect depends on the **selector's own Hessian estimate** (§5.3), and a **model-scope boundary**: two of four checkpoints exhibit the collapse, and on Qwen2.5-3B no selector is harmful at all (§5.4).
6. Supporting audits under matched storage with random controls, including a matched-bit downstream signal-versus-random test (§7).

Scope of the claim. We do not claim perplexity is uninformative, nor that this configuration is one practitioners would deliberately choose. We claim that a perplexity gap of this magnitude, produced by an ordinary PTQ manipulation, is not sufficient evidence of utility loss—and that quantization studies and regression gates relying on perplexity alone can therefore be badly miscalibrated.

What is not new. Activation-outlier protection [Dettmers et al., 2022], Hessian- and error-based column selection [Lee et al., 2024, Dettmers et al., 2023], and integrating protected-column selection into the compensation pass [Lee et al., 2024] are all prior work. Our contribution is the controlled decoupling demonstration, the structural condition on the selector, and the negative mechanism result.

2 Related work

Outlier-aware mixed precision. Dettmers et al. [2022] decompose the matmul so outlier feature dimensions stay in FP16. OWQ [Lee et al., 2024] scores input columns by Hessian-weighted quantization error, keeps the top ones in FP16, and—crucially for our analysis—orders them last inside the OPTQ pass, so compensation proceeds knowing which columns remain exact. SpQR [Dettmers et al., 2023] isolates outlier weights in a sparse high-precision format. Atom [Zhao et al., 2024] and SqueezeLLM [Kim et al., 2023] use related group/column/row schemes; AWQ [Lin et al., 2024] scales channels by activation statistics. Our selectors are drawn from this family; we vary the signal and the base rather than proposing a new protector.

Error compensation. GPTQ [Frantar et al., 2023], building on OBQ [Frantar and Alistarh, 2022], quantizes column by column and propagates each column's rounding error into the not-yet-quantized columns, minimizing $\|\Delta W X\|^2$. The residual it leaves is therefore structured, which is what makes the base$\times$selector interaction we study non-trivial. HQQ [Badri and Shaji, 2023] is calibration-free but is *not* plain round-to-nearest: it optimizes quantization parameters with a half-quadratic splitting procedure under a robust reconstruction loss.

selector	ΔW	H diag	H off-diag
`random` (3 seeds)	—	—	—
`act_max`, `act_scale`	—	—	—
`residual_max/rms`	✓	—	—
`hessian_diag`	—	✓	—
`greedy_indep`	✓	✓	✓
`greedy` (iterative)	✓	✓	✓

Table 1: Selection signals and which parts of the objective each uses.

Evaluating quantized models. Broad benchmark studies [Jin et al., 2024] report downstream task results for quantized LLMs, and several works note that perplexity and task metrics can move differently. Our contribution is sharper and adversarial: rather than observing correlation strength across methods, we *construct* a checkpoint with an extreme perplexity gap and verify that the downstream cost is nil, which bounds how much a perplexity delta can be trusted.

3 Setup

Protection form. For a linear layer with weight $W \in \mathbb{R}^{o \times i}$ and protected input-channel set S, the protected forward pass is $y = Q(W)x + x[S] \cdot (W - Q(W))[:, S]^\top$. Materializing gives $W_{\text{dense}} = Q(W) + \text{scatter}(W - Q(W), S)$, algebraically identical to the forward pass, so a faithful export must reproduce the runtime perplexity. We use this identity as an integrity check (§4.2). Storage is additive: the low-bit base is retained and FP16 correction columns are stored on top, rather than replacing base columns.

Bases. (i) **HQQ-4** [Badri and Shaji, 2023], calibration-free, no error compensation. (ii) **GPTQ-4** [Frantar et al., 2023], error-compensated, produced at W4 group size 128 with a fixed WikiText-2 calibration set (128×2048 tokens), imported as fake-quantized weights and re-evaluated in our own evaluator.

Selectors. With residual $\Delta W = W - Q(W)$ and input Hessian $H = X^\top X$, Table 1 lists the signals. `greedy_indep` is a *one-shot full-Hessian* selector, not an activation-magnitude score. The last two maximize the exact reduction in $\|\Delta W X\|^2 = \mathrm{tr}(\Delta W H \Delta W^\top)$ obtainable by restoring columns.

Storage. We report **matched theoretical weight-only storage**: bits per quantized-linear parameter, counting low-bit values, group scales/zeros at the base's true group size (128 for the imported GPTQ base), FP16 correction columns and channel indices. Embeddings, `lm_head` and norms are FP16 in every method and excluded from the axis; all runs exclude `lm_head` from quantization so the quantized scope matches the baseline exactly. No kernels, packing or latency are modelled.

Evaluation. WikiText-2 [Merity et al., 2017] perplexity, sequence length 2048, canonical non-overlapping chunking, one evaluator for every perplexity in this paper; exported checkpoints are evaluated after reload. Downstream: lm-evaluation-harness [Gao et al., 2024] on six zero-shot tasks—HellaSwag [Zellers et al., 2019], ARC-easy and ARC-challenge [Clark et al., 2018], PIQA [Bisk et al., 2020], WinoGrande [Sakaguchi et al., 2021] and LAMBADA [Paperno et al., 2016]—on full test sets, with paired-bootstrap 95% confidence intervals on per-example correctness. The imported GPTQ base measures 10.557 on Llama-3.2-1B in our evaluator versus 10.363 as reported by the producing toolkit, so we use the **reloaded** $k = 0$ **checkpoint** as the operational base for every contrast, keeping all comparisons inside one evaluator.

point	WT2 PPL	macro acc	LAMB. PPL
FP16	7.817	68.43%	—
GPTQ-4 (base)	8.172	67.10%	4.28
greedy@GPTQ, 2%	**51.68**	**67.61%**	**4.17**
resmax@GPTQ, 2%	8.100	67.47%	—
HQQ-4 (base)	8.328	66.86%	—
best greedy@HQQ, 20%	7.994	68.14%	—
random@HQQ, 20%	8.247	67.21%	—

Table 2: Llama-3.2-3B operating points. Accuracy is `acc_norm` where the task provides it and `acc` otherwise, macro-averaged over six tasks, from the same artifact the confidence intervals are computed from.

contrast	Δ (pts)	95% CI
greedy@GPTQ − GPTQ-4	$+0.51$	$[+0.13, +0.86]$
resmax@GPTQ − GPTQ-4	$+0.37$	$[-0.06, +0.77]$
best@HQQ − HQQ-4	$+1.28$	$[+0.84, +1.68]$
best@HQQ − random@HQQ	$+0.92$	$[+0.50, +1.34]$

Table 3: Paired-bootstrap macro-Δ accuracy, Llama-3.2-3B.

Calibration provenance. The base quantizer's Hessian came from $128 \times 2048 \approx 262$k WikiText-2 tokens. Our selectors' Hessian, by default, came from a 51-prompt instruction set of $\approx$500 tokens—rank-deficient for layers with 3072–8192 input channels, and a different distribution. We treat this as an experimental variable rather than a fixed detail, and measure its effect in §5.3.

4 A perplexity collapse with no downstream cost

4.1 The two measurements

On Llama-3.2-3B with the GPTQ-4 base (8.172 in our evaluator, 4.25 bits), protecting 2% of input channels with `greedy` gives 51.68 perplexity at 4.57 bits. Entries in Table 2 share identical storage at a given budget.

The model whose perplexity rose $6.3\times$ is *not* worse downstream; its confidence interval against its own base excludes zero on the favourable side. Its LAMBADA perplexity—a token-level metric on a different corpus—is 4.17, slightly better than the base's 4.28. Figure 1 shows the whole picture.

4.2 The checkpoint is the same model

Because this is the crux, we establish checkpoint identity explicitly. The protection form guarantees $\mathrm{PPL}_{\mathrm{runtime}} = \mathrm{PPL}_{\mathrm{materialized}}$ algebraically (§3), and we confirm it empirically at each stage of the chain that produced the downstream number: in-sweep runtime 51.68; in-memory materialization 51.76; the exported checkpoint reloaded from disk **51.76** (PASS, tolerance 0.5); the same directory scored by lm-evaluation-harness, macro **67.61%**. The reload was run on the exact directory the harness loaded, after the downstream evaluation.

It replicates on a second model. Llama-3.2-1B reproduces the pattern under the same verification discipline (Table 4): the exported checkpoint reloads at **63.83** against a sweep value of 63.95 (PASS, $\Delta = 0.124$), a $6.0\times$ increase over its 10.557 base, and scores macro **58.67%** against the base's **58.35%**. The interval

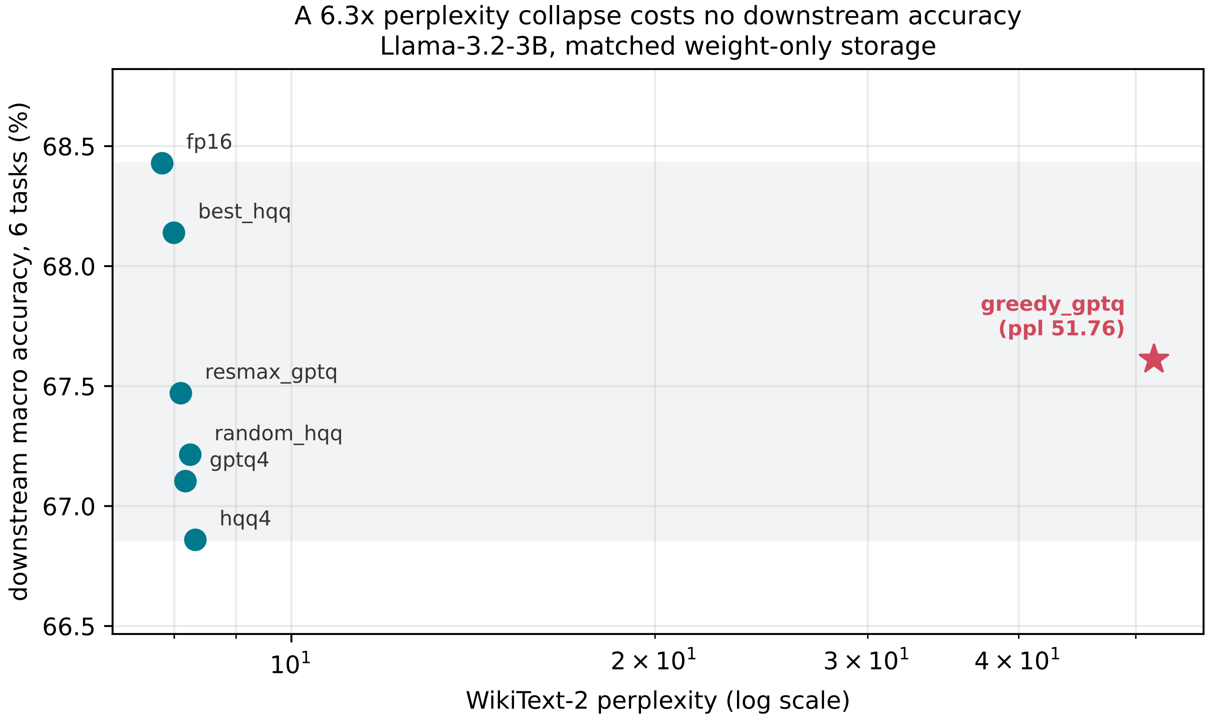

Figure 1: Every operating point of Llama-3.2-3B: WikiText-2 perplexity (log) against downstream macro accuracy. The catastrophic point sits $6.3\times$ to the right yet at the same height, above both the GPTQ-4 and HQQ-4 bases.

model	base	greedy (reload)	macro Δ
3B	8.172	51.76 (PASS)	$+0.51$ [$+0.13$, $+0.86$]
1B	10.557	63.83 (PASS)	$+0.32$ [-0.07, $+0.73$]

Table 4: Both susceptible models: a $\sim 6\times$ perplexity collapse with no measurable downstream cost.

includes zero, so on 1B we claim no degradation rather than an improvement; on 3B the interval excludes zero on the favourable side.

4.3 The damage is broad, not a tail artifact

The obvious explanation is that perplexity, being $\exp$ of a *mean* negative log-likelihood, is dominated by its worst tokens: if a small minority of tokens became catastrophically improbable, perplexity would explode while the bulk of the model's behaviour stayed intact. We pre-registered this hypothesis and tested it directly, scoring both checkpoints on the identical token stream (141 windows, 288,627 supervised targets) and decomposing the per-token negative log-likelihood.

The hypothesis is false. The degradation is broad: **74.6%** of tokens have higher NLL, 25.4% lower, and the median ΔNLL is $+0.287$. The worst 1% of tokens carry only 6.9% of the total increase, and the worst 10% only 46.3%. Table 5 gives the perplexity after excluding the worst-k% of tokens for both models.

The *median* token is meaningfully worse and three quarters of all tokens degrade. This is not a small set of outliers dragging up a mean; the model is genuinely and broadly worse at next-token prediction on

excluded	base	greedy@GPTQ
0%	8.17	51.76
1%	8.30	46.85
5%	8.59	**34.56**

Table 5: Trimmed perplexity, Llama-3.2-3B. Removing the worst 5% of tokens—fifty times more than a tail explanation would require—still leaves a $4\times$ gap.

selector	Hessian use	3B	1B
`random`	none	8.181	10.41
`act_max`	none	8.181	10.71
`act_scale`	none	8.586	10.93
`residual_max`	none	8.100	10.42
`residual_rms`	none	8.146	10.43
`hessian_diag`	diag only	**8.153**	**10.441**
`greedy_indep`	full, one-shot	41.50	11.28
`greedy`	full, iterative	**51.68**	**63.95**

Table 6: All selectors at 2%, GPTQ-4 base (3B base 8.172; 1B base 10.557).

WikiText-2. That makes the result more surprising, not less, and it removes the most comfortable way to dismiss it. A model can be substantially and pervasively worse at modelling a corpus while answering six downstream benchmarks exactly as well as before. We do not have a positive account of why, and we do not offer one.

5 What generates the perplexity damage

5.1 The damage requires off-diagonal Hessian coupling

Table 6 gives all selectors at matched storage on the GPTQ-4 base at 2% protection. `hessian_diag`—which uses the Hessian, but only its diagonal—is benign at every budget we tested (3B: 8.153/8.163/8.156/8.147 at 2/5/10/20%; 1B: 10.441/10.462/10.475/10.450). Only the two selectors using the **off-diagonal**, cross-column terms cause damage. This rules out both "reads the quantization residual" and "uses the Hessian" as the operative property and isolates the interaction structure (Figure 2).

We avoid absolutes: the benign selectors are not exactly neutral. `act_scale` reaches 8.586 at 2% on 3B and `random` reaches 9.048 at the 10% budget—moderate movements, one to two orders of magnitude smaller than the collapse. Throughout we call a change *indistinguishable* within ± 0.1 PPL of the base, *moderate* up to $\sim$1 PPL, and *catastrophic* above $5\times$ the base.

5.2 Severity within the coupled selectors

Among the two coupled selectors, iterative re-scoring is at least as harmful as one-shot evaluation, but the separation is model-dependent and not monotone in budget. On 1B the ordering is clean at 2% (63.95 vs 11.28); on 3B both collapse (51.68 vs 41.50) and `greedy` is worse at 2% and 5% but *less* harmful at 10% and 20% (38.39, 40.65 vs 44.83, 43.04). We therefore report that coupling is necessary for harm, and that harm saturates rather than scaling smoothly once present.

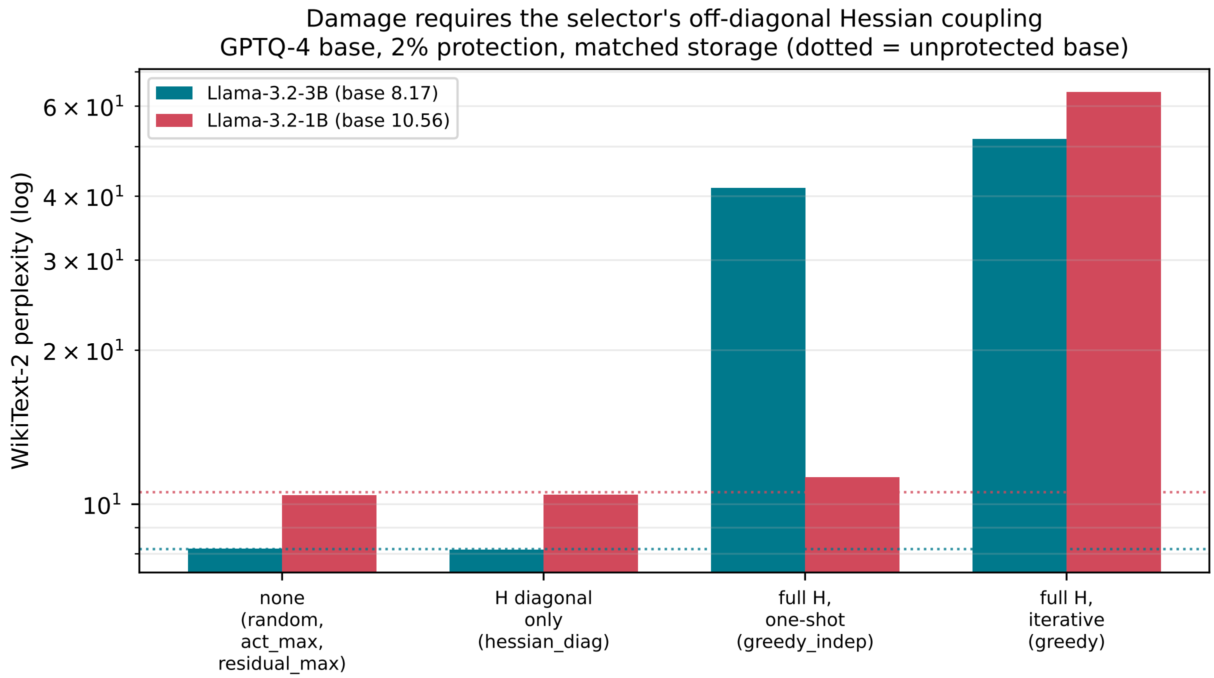

Figure 2: Perplexity by how much of the Hessian objective the selector uses. Dotted lines are the unprotected bases.

5.3 The selector's own Hessian estimate matters

Our default selector Hessian used ≈500 tokens of out-of-distribution prompts, against 262k in-distribution tokens for the base. Re-running `greedy` with the selector Hessian estimated from the same 128×2048 WikiText-2 tokens as the base gives 3B $51.68 \rightarrow$ **58.24** and 1B $63.95 \rightarrow$ **11.39**. On 3B the collapse persists and is slightly worse, so it is **not** an artifact of an under-estimated Hessian. On 1B most of it disappears (11.39 against a 10.557 base—moderate, not catastrophic), so on that model the estimate quality was doing much of the work. Any study using Hessian-based selection should report the selector's calibration size and distribution; we did not initially, and it materially changes one of our two models.

5.4 On which models the collapse occurs at all

The collapse is not universal. Running the identical configuration across four checkpoints: Llama-3.2-1B ($10.557 \rightarrow 63.95$) and Llama-3.2-3B ($8.172 \rightarrow 51.68$) collapse; Qwen2.5-3B does not ($8.290 \rightarrow 8.335$, i.e. $+0.045$); Llama-2-7B does not (5.571 against an FP16 reference of 5.469; its unprotected base was not measured in that run, so we treat it as weaker evidence).

On Qwen2.5-3B the *entire* selector panel is benign at 2%—`greedy` 8.335, `greedy_indep` 8.357, `residual_rms` 8.339, `residual_max` 8.351, `act_max` 8.366, `act_scale` 8.372, `random` 8.300, against a base of 8.290. The spread across every selector is under 0.09 perplexity, so on this model the choice of selection signal is immaterial.

The collapse therefore requires the structural condition of §5.1 *and* a susceptible model. Two of four checkpoints are, both from the same family. We have no predictor for susceptibility, and finding one is the clearest route to turning this boundary into a diagnostic. Note that these models cannot supply further decoupling replication: a model healthy in perplexity has nothing to decouple, so evaluating it downstream

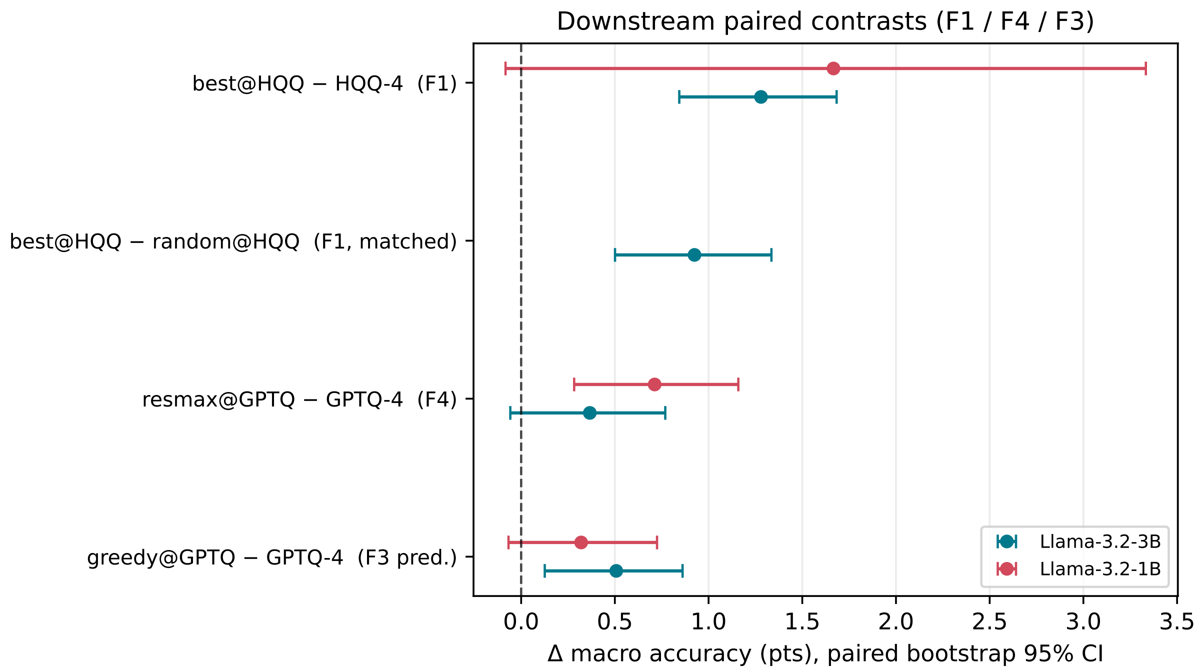

Figure 3: Downstream paired contrasts with 95% confidence intervals.

would only confirm that a healthy model scores healthily.

6 A falsified mechanism

The natural explanation for §5 is that the coupled selectors identify columns carrying GPTQ's deposited error correction, and that restoring those columns to FP16 discards compensation the remaining columns were tuned around. This predicts that harmful selectors should rank compensation-bearing columns highly.

We tested it directly. During a sequential GPTQ pass we recorded, per input channel, the compensation magnitude $\mathrm{comp}_j = \|W^{\text{pre-quant}}_{:,j} - W^{\text{orig}}_{:,j}\|_2$ —how far compensation moved that column before it was itself quantized—and rank-correlated each selector's score against it (Spearman, median over 56 layers per model): `residual_rms` $+0.988$ (3B) / $+0.980$ (1B); `residual_max` $+0.545$ / $+0.614$; `greedy_gain` $+0.390$ / $+0.386$; `hessian_diag` -0.134 / -0.083.

The prediction fails, and fails informatively: the selector most nearly *identical* to compensation magnitude (`residual_rms`, $\rho \approx 0.99$) is among the safest, while the catastrophic selector is only moderately aligned. Protecting compensation-bearing columns is therefore not sufficient—and evidently not necessary—for the collapse. We report this as a negative result and do **not** substitute a new mechanism: §5 establishes a structural condition, not a causal account.

7 Supporting audits

Protection helps a calibration-free base. On HQQ-4, every informative signal beats the matched-bit random control at both budgets and both sizes (3B at 2%: `greedy` 8.100, `act_max` 8.108, `residual_rms` 8.141 vs `random` 8.319). Downstream this reproduces at matched storage: best@HQQ − random@HQQ $= +0.92$ pts $[+0.50, +1.34]$ (Figure 3), so the gain is a selection effect, not a budget effect.

Interaction modelling does not pay where protection helps. On HQQ, iterative and one-shot selection are within ≈ 0.03 PPL at every budget—we observed no material difference, which is not an equivalence result—while on the compensated base the iterative variant is strictly worse.

The base dominates the frontier. At matched weight-only storage the base quantizer sets the ceiling: the GPTQ-4 base at 4.25 bits (3B 8.172) is not reached by any protected HQQ configuration at comparable storage, and the best protection on GPTQ buys ≈ 0.07 PPL (`residual_max`, 8.100 at 4.57 bits). We state this as an observation on this grid, not a general law.

8 Conclusion

We constructed, verified and evaluated checkpoints whose WikiText-2 perplexity is $\sim 6\times$ their base's and whose downstream accuracy is not worse—on 3B slightly better, with a confidence interval excluding zero—across six zero-shot tasks, on two models, with a token-level metric on a second corpus also unharmed. The construction uses only standard PTQ components. A per-token decomposition shows the perplexity damage is broad rather than a tail artifact: three quarters of tokens degrade and the median token worsens, so the models really are pervasively worse at modelling the corpus while answering the benchmarks exactly as well. We further show that producing this damage requires the selector's off-diagonal Hessian coupling, and that the obvious mechanism—destroying error compensation—is falsified by direct measurement.

The practical consequence is narrow and concrete. Perplexity remains a useful diagnostic, but a perplexity delta, even a very large one, is not by itself evidence of utility loss for a quantized model. Quantization studies should report task metrics alongside perplexity, and perplexity-only regression gates should be expected to reject models that are downstream-equivalent.

Limitations

Scale and family. The decoupling is verified on both susceptible checkpoints we have, Llama-3.2-3B and 1B, but both come from one model family. Of the four checkpoints tested only these two exhibit the perplexity collapse at all (§5.4), so Qwen2.5-3B and Llama-2-7B cannot supply further replication. Broadening the claim requires finding further *susceptible* models, which in turn requires a predictor for susceptibility that we do not have.

No causal mechanism, and no positive account of the decoupling. §6 falsifies the compensation explanation and §4.3 falsifies the tail explanation; §5 gives a structural condition only. We can say what the decoupling is *not* caused by more confidently than what it is caused by, and a numerical-conditioning explanation for the collapse is not excluded.

The ordering intervention did not produce a usable result. Our internal sequential GPTQ yields a degenerate base at 3B (perplexity 3437 unprotected, against an FP16 reference of 7.82), so all three arms are uninterpretable and we exclude the experiment entirely rather than report its arms. The causal test of whether ordering matters therefore remains open.

Task battery. Six zero-shot tasks, five of them multiple-choice. A perturbation invisible here could still harm long-form generation, instruction following or reasoning chains; our claim is about the *inference from perplexity*, not a guarantee of preserved capability.

Other. Our default selector Hessian was small and out-of-distribution (§5.3). Storage is theoretical weight-only bytes with an additive representation; no kernels or latency are modelled. Downstream confidence intervals are paired bootstraps over examples and do not capture variance over random channel-selection sets, for which we ran a single seed downstream (three in the perplexity sweeps).

Ethical considerations

Reliable quantization lowers the memory, energy and hardware cost of deploying language models, widening access. The failure mode we document cuts both ways: a perplexity-only gate may reject a usable model, wasting effort, and—more importantly—the converse inference is equally unsafe, so small perplexity movements should not be read as evidence that a quantized model is safe to deploy. Practitioners with limited evaluation budgets are the most likely to rely on a single perplexity number, and are therefore the most exposed. Our results are benchmark findings and are not guarantees of behaviour in safety-critical deployments. The work uses publicly available pretrained models and benchmark corpora, and involves no human subjects or personal data.

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A Provenance

Every perplexity is read from a committed JSON artifact carrying its own `skip_lm_head`, `base_group_size`, `seed`, `selector_calib_source` and `selector_hessian_tokens` fields. Five values are recorded for Llama-3.2-3B `greedy@GPTQ` at 2%, from distinct runs: 51.68 (corrected matrix sweep), 51.76 (disk reload of the exported checkpoint), 55.35/55.19 (standalone diagnostic, runtime and materialized), and 58.24 (selector Hessian matched to the base). All are catastrophic relative to the 8.172 base. Tables use the matrix-sweep value except §5.3, which reports the matched-calibration run explicitly.

Llama-3.2-1B. An earlier export of `greedy@GPTQ` reloaded at 203.72 against a sweep value of 63.95 and was excluded. It was re-exported from scratch and now reloads at 63.83 (PASS, $\Delta = 0.124$); §4.2 uses the repaired checkpoint. The discarded export is retained in the artifacts for audit.

Excluded experiment. The ordering intervention ran to completion but on a degenerate internal base: unprotected sequential-GPTQ perplexity 3437.3 against an FP16 reference of 7.82. Its three arms (3437.3 / 3434.8 / 3448.6) are mutually indistinguishable because all are broken, not because ordering is irrelevant, so no conclusion is drawn from them.

B Statistics

Random controls. `random` is run with three seeds per (model, base, budget) in the perplexity sweeps; tables report the seed-1234 value. Downstream, `random@HQQ` was evaluated at one seed.

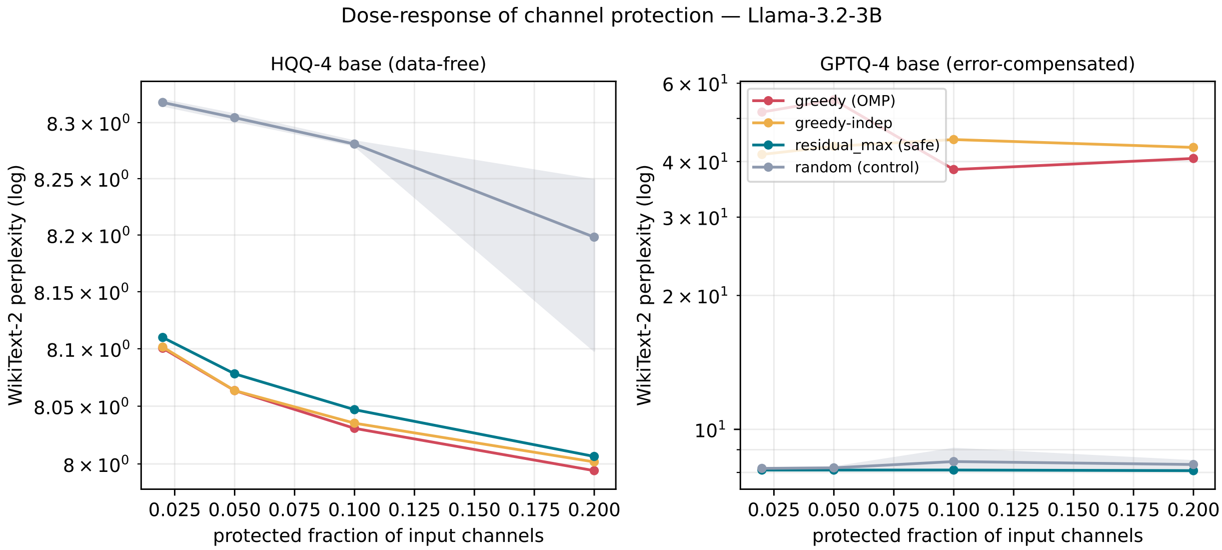

Figure 4: Perplexity against protection budget, Llama-3.2-3B, both bases.

Downstream confidence intervals. The harness is run with `-log_samples`, giving per-example correctness. For a contrast we pair correctness on the same examples and bootstrap the mean difference (2000 resamples); the macro difference bootstraps within each task and averages.

Storage formula. Weight-only bits per quantized-linear parameter: low-bit values + group scales/zeros at the base's true group size + FP16 correction columns + channel indices; embeddings, `lm_head` and norms excluded and FP16 in every method.

Dose–response. Figure 4 gives the full budget sweep on both bases.